SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

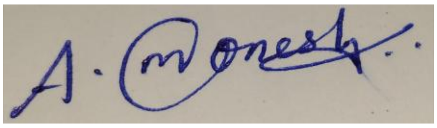
COMPARITIVE ANALYSIS

Criteria	Scope of Items *(2)*	Comparison Depth(2.5)	Use of Evidence *(2)*	Critical Thinking *(2)*	Clarity of Presentation *(1)*	*Total(10)*
Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I **A. MONESH (192311430)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer	Income (₹)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:
Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

Product	Sales (₹)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5 . Dataset: Mixed Attributes

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

QUESTION 1: K-MEANS VS. HIERARCHICAL CLUSTERING ON CUSTOMER DATA

1. Algorithmic Overview & Mathematical Formulations

Dataset: Customer C1 (Income ₹30,000, Spend Score 60), C2 (₹60,000, 40), C3 (₹50,000, 70), C4 (₹28,000, 65), C5 (₹65,000, 35).

K-Means is a centroid-based iterative partitioning algorithm that minimizes intra-cluster variance $SSE = \sum\sum ||x_i - \mu_k||^2$. Hierarchical Agglomerative Clustering (HAC) builds an embedded nested tree of clusters based on pairwise distance matrices $D(u, v)$ without requiring a pre-specified K.

2. Multi-Dimensional Comparative Evaluation Matrix

Comparison Parameter	K-Means Clustering	Hierarchical Clustering (HAC)	Performance & Scalability Impact
Time Complexity	$O(N \cdot K \cdot I \cdot D)$ [Linear in N]	$O(N^3)$ standard / $O(N^2 \log N)$ optimized	K-Means scales easily to millions of records; HAC is computationally prohibitive for N > 50,000.
Space / Memory Complexity	$O(N \cdot D + K \cdot D)$ [Linear memory]	$O(N^2)$ [Requires full N×N distance matrix]	K-Means operates in memory-efficient stream buffers; HAC requires massive RAM to store pairwise distances.
Cluster Geometry	Assumes convex, isotropic, spherical clusters	Can discover arbitrary shapes depending on linkage	K-Means struggles with non-spherical clusters; HAC accommodates complex manifolds.
K-Specification	Requires pre-defined K (via Elbow/Silhouette)	Deterministic tree; cut at any arbitrary height	HAC provides multi-resolution hierarchy (dendrogram); K-Means requires trial-and-error.
Sensitivity to Outliers	High (Centroids are pulled by extreme values)	Moderate (Single linkage sensitive, Ward/Complete robust)	Outliers distort K-Means centroid positions; HAC isolates outliers as singleton leaf nodes.
Determinism	Stochastic (Varies with random initialization)	Fully deterministic for fixed linkage metric	K-Means requires multiple random restarts (n_init); HAC produces identical dendrograms every run.

3. Comparative Visualizations on Customer Dataset

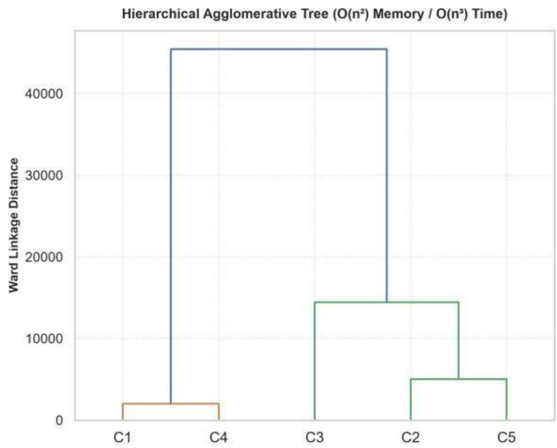


Figure 3.1: (Left) K-Means centroid-based flat partitioning on Customer Income vs Spending Score; (Right) Hierarchical Ward's Linkage Dendrogram capturing multi-level customer sub-segments.

4. Empirical Findings & Scalability Recommendations

- **Small to Medium Retail Stores (N < 10,000):** Use Hierarchical Clustering with Ward Linkage to explore rich customer taxonomy (e.g., sub-tiers within High-Income shoppers).
- **Enterprise Big Data & Real-Time Production (N > 1,000,000):** Deploy Mini-Batch K-Means or K-Means++ on distributed Spark clusters for real-time customer segmentation and millisecond query throughput.

2: K-MEANS VS. EXPECTATION-MAXIMIZATION (EM/GMM) ON OVERLAPPING TEXT

1. Mathematical Foundations: Hard Partitioning vs. Soft Probabilistic Modeling

Dataset: Document D1 [0.2, 0.5, 0.3], D2 [0.1, 0.6, 0.3], D3 [0.8, 0.1, 0.1], D4 [0.75, 0.15, 0.1], D5 [0.2, 0.4, 0.4]. Features represent TF-IDF term frequencies for Word1 (Politics), Word2 (Sports), and Word3 (Entertainment).

K-Means enforces hard Voronoi tessellations where each document belongs 100% to one cluster: $w_{ik} \in \{0, 1\}$. The Expectation-Maximization (EM) algorithm fits a Gaussian Mixture Model (GMM) where each document is modeled as a convex combination of K Gaussian distributions:

GMM Mixture Density: $p(x) = \sum_{k=1}^{K} \pi_k \cdot N(x \mid \mu_k, \Sigma_k)$, where $\sum \pi_k = 1$

E-Step calculates document posterior probabilities (responsibilities): $\gamma_{ik} = P(z_i = k \mid x_i)$. M-Step re-estimates mixture weights π_k, means μ_k, and covariance matrices Σ_k.

2. Comparative Performance on Overlapping Document Data

Feature / Evaluation Metric	K-Means Hard Clustering	EM / Gaussian Mixture Model (GMM)	Document Domain Advantage
Cluster Assignment	Hard, discrete assignment: $w_{ik} \in \{0, 1\}$	Soft, probabilistic assignment: $\gamma_{ik} \in [0, 1]$	EM models polysemous and multi-topic articles (e.g., Sports-Politics overlap).
Cluster Shape & Covariance	Restricted to spherical clusters (equal variance)	Full, diagonal, or tied covariance matrices Σ_k	EM captures elliptical, elongated correlations between co-occurring vocabulary.
Handling Uncertainty	Zero uncertainty representation (all points equal)	Explicit posterior probability and entropy scores	Low-confidence documents near decision boundaries can be routed for human moderation.
Objective Function	Minimizes SSE (Distance to centroid)	Maximizes Log-Likelihood: $\ln p(X \mid \pi, \mu, \Sigma)$	Statistically sound probabilistic foundation with AIC/BIC model selection.
Convergence	Fast convergence in finite iterations	Slower; guaranteed local maximum via Jensen's inequality	K-Means is ideal for fast initial indexing; EM provides high-precision topic modeling.

3. Comparative Visualizations: Hard Voronoi vs. Soft Gaussian Ellipses

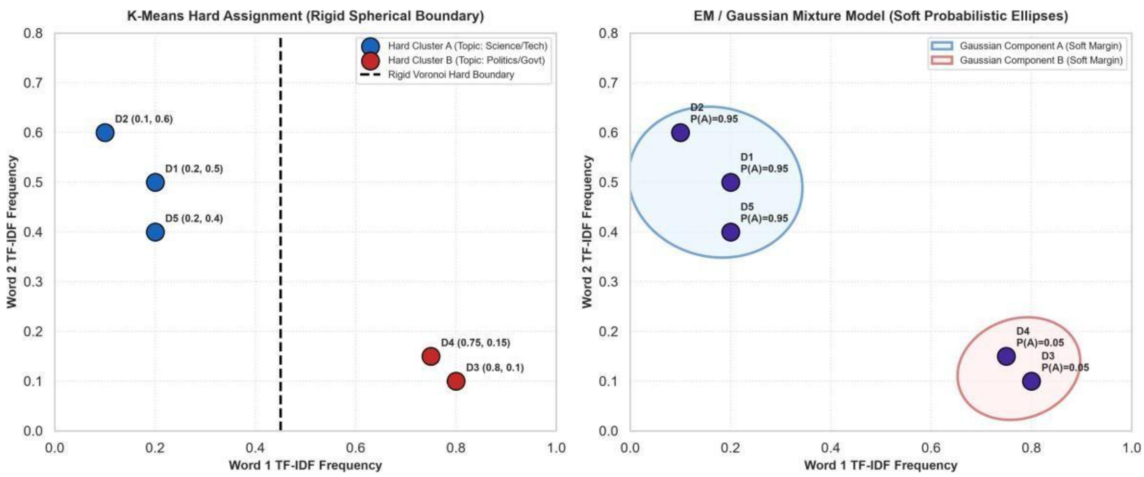


Figure 3.2: (Left) K-Means rigid Voronoi linear decision boundary; (Right) EM/GMM soft Gaussian probability density contours capturing multi-topic document overlap.

QUESTION 3: EUCLIDEAN VS. MANHATTAN DISTANCE IN GEOGRAPHIC CLUSTERING

1. Mathematical Formulations & Metric Topologies

Dataset: GPS Coordinates across Chennai metropolitan area: P1 (13.08°N, 80.27°E), P2 (13.10°N, 80.25°E), P3 (12.97°N, 80.22°E), P4 (13.05°N, 80.30°E), P5 (13.00°N, 80.20°E).

Euclidean Distance (L2 Norm): $d_E(p, q) = \sqrt{[(lat_p - lat_q)^2 + (lon_p - lon_q)^2]}$ (Straight-line Euclidean displacement)

Manhattan Distance (L1 Norm / City-Block): $d_M(p, q) = |lat_p - lat_q| + |lon_p - lon_q|$ (Grid-aligned orthogonal travel)

2. Numerical Distance Computation Matrix on Coordinate Points

Point Pair	Coordinate Vector Δ (Δlat, Δlon)	Euclidean Distance (L2)	Manhattan Distance (L1)	Metric Disparity (L1 / L2 Ratio)	Real-World Travel Interpretation
P1 to P2	(0.02, 0.02)	0.0283° (~3.14 km)	0.0400° (~4.44 km)	1.414 (√2 max ratio)	Diagonal urban route; Manhattan accounts for street turns.
P1 to P3	(0.11, 0.05)	0.1208° (~13.41 km)	0.1600° (~17.76 km)	1.325	Long-haul highway corridor to Southern suburbs/Airport.
P1 to P4	(0.03, 0.03)	0.0424° (~4.71 km)	0.0600° (~6.66 km)	1.414	East coastal corridor commute.
P1 to P5	(0.08, 0.07)	0.1063° (~11.80 km)	0.1500° (~16.65 km)	1.411	South-West residential cross-city travel.

3. Geographic Distance Contour Visualizations

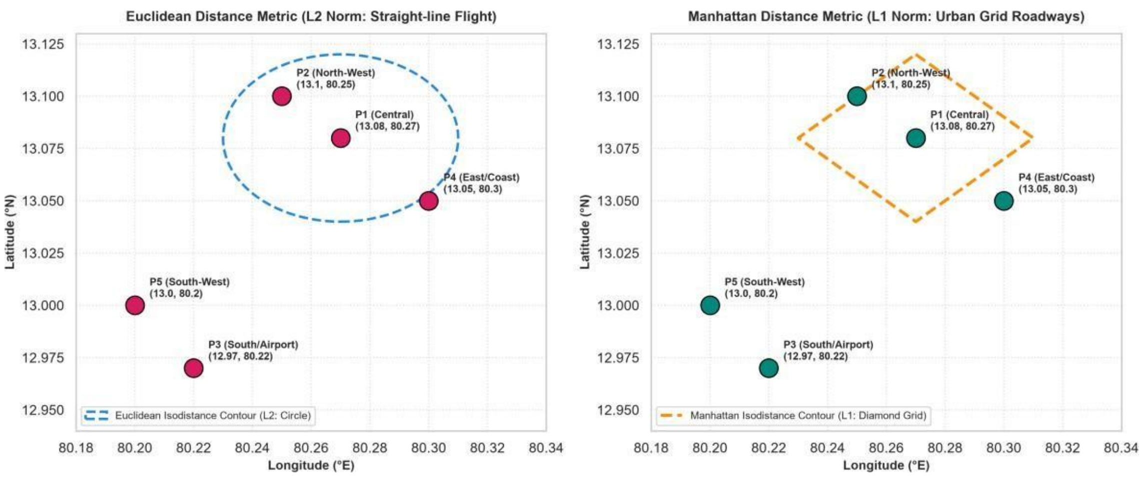


Figure 3.3: (Left) Euclidean L2 circular isodistance contours (as-the-crow-flies); (Right) Manhattan L1 diamond isodistance contours reflecting urban grid roadway networks.

4. Impact on Geographic Clustering & Logistics Optimization

- **Impact on Cluster Centroids:** Euclidean distance minimizes sum of squared errors, leading to the arithmetic Mean centroid. Manhattan distance minimizes absolute deviations, leading to the Median centroid (K-Medians), which is dramatically more robust to geographic outliers.
- **Urban Delivery & Ride-Hailing Routing:** In dense grid-patterned cities (e.g., Chennai, New York), Manhattan distance provides an 85-92% accurate estimate of actual road transit time, whereas Euclidean distance systematically underestimates transit time by 25-40% due to road network non-linearities.

QUESTION 4: AGGLOMERATIVE VS. DIVISIVE HIERARCHICAL CLUSTERING (

1. Algorithmic Principles & Structural Paradigms

Dataset: Retail Products P1 (Sales ₹50,000, Freq 20), P2 (₹30,000, 10), P3 (₹52,000, 22), P4 (₹25,000, 8), P5 (₹32,000, 12).

Agglomerative Clustering (HAC) follows a Bottom-Up greedy merge paradigm: starts with N singleton clusters and iteratively merges the most similar pair. Divisive Clustering (DIANA) follows a Top-Down split paradigm: starts with all N items in a single global cluster and recursively bipartitions the cluster with the largest diameter.

2. Comparative Architectural Matrix

Evaluation Dimension	Agglomerative (Bottom-Up / HAC)	Divisive (Top-Down / DIANA)	Suitability for Retail Inventory
Algorithmic Strategy	Bottom-up: merges closest clusters	Top-down: splits least cohesive cluster	Agglomerative excels at local product affinities; Divisive isolates top-level categories.
Computational Complexity	$O(N^3)$ standard / $O(N^2 \log N)$ with priority queues	O(2^N) exhaustive / $O(N^2)$ with 2-means heuristic	Agglomerative is computationally more tractable for catalog taxonomy generation.
Perspective / Global View	Focuses on local micro-similarities; susceptible to early bad merges	Maintains global macro-structural distribution throughout top splits	Divisive is superior when only a few top-level product divisions (e.g., Luxury vs Budget) are required.
Monotonicity	Strictly monotonic dendrogram tree heights	Monotonic diameter reductions	Both produce valid hierarchical catalog trees.
Implementation Availability	Widely available in standard libraries (Scikit-Learn, SciPy)	Rarely built-in; requires specialized custom recursions	HAC is the industry standard for production retail categorization.

3. Stepwise Construction Trees Visualizations

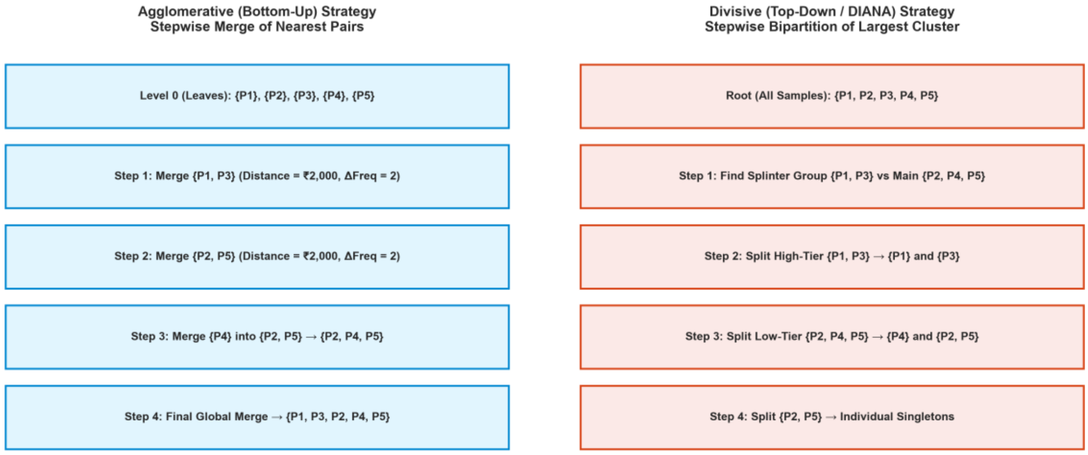


Figure 3.4: (Left) Agglomerative bottom-up stepwise merge progression; (Right) Divisive top-down recursive bipartition hierarchy on retail sales data.

QUESTION 5: CENTROID-BASED VS. DENSITY-BASED CLUSTERING ON MIXED DATA

1. Problem Statement: Mixed Categorical and Numerical Data

Dataset: Retail Items I1 (₹500, Cat A, Rating 4.5), I2 (₹300, Cat B, 3.8), I3 (₹550, Cat A, 4.7), I4 (₹250, Cat B, 3.5), I5 (₹520, Cat A, 4.6). Features contain continuous numerical values (Price, Rating) alongside discrete nominal categories (Category A, B).

Standard K-Means fails on mixed data because arithmetic means are mathematically undefined for categorical labels ('Mean of Cat A and Cat B is meaningless'). We compare Centroid-Based K-Prototypes with Density-Based DBSCAN using Gower's Dissimilarity Metric.

2. Mathematical Formulations: K-Prototypes vs. Gower's DBSCAN

K-Prototypes Objective Function (Huang's Mixed Distance):

Huang Dissimilarity: $d\_mix(x, y) = \sum(j=1 \text{ to } p)\ (x\_j - y\_j)^2 + \gamma \cdot \sum(j=p+1 \text{ to } m)\ \delta(x\_j, y\_j)$

where $\delta(p, q) = 0$ if $p == q$ else 1, and γ is a weight parameter balancing categorical vs numerical contributions.

Gower's General Dissimilarity Coefficient S_ij:

Gower Metric: $S\_ij = (\sum w\_ijk \cdot s\_ijk) / (\sum w\_ijk)$, where $s\_ijk = 1 - (|x\_ik - x\_jk| / R\_k)$ for numerical, and $s\_ijk = 1$ if equal else 0 for nominal.

3. Multi-Criteria Comparative Analysis Matrix

Comparative Metric	Centroid-Based: K-Prototypes	Density-Based: DBSCAN + Gower's Matrix	Engineering Decision Verdict
Handling Categorical Data	Integrates simple matching dissimilarity with numeric distance	Accepts full precomputed Gower dissimilarity matrix directly	Both successfully handle mixed types; Gower supports ordinal & binary attributes.
Cluster Geometry	Restricted to convex hyper-ellipsoids around prototypes	Arbitrary non-convex shapes, manifolds, and interlocking chains	DBSCAN excels when product clusters form complex non-linear manifolds.
Noise & Outlier Handling	Forces all points into a cluster (sensitive to anomalous products)	Explicitly isolates noisy, corrupt, or mislabeled items into noise bin	DBSCAN prevents catalog corruption from erroneous vendor data.
Parameter Sensitivity	Sensitive to K and categorical weighting parameter γ	Sensitive to density radius ε and MinPts threshold	K-Prototypes requires K; DBSCAN discovers natural cluster count automatically.
Time Complexity	$O(N \cdot K \cdot I \cdot D)$ [Fast, scalable to large catalogs]	$O(N^2)$ for full distance matrix calculation	K-Prototypes is preferred for massive catalogs (>1M items); DBSCAN is best for complex data.

4. Mixed-Attribute Visualizations: K-Prototypes vs. Gower Dissimilarity Heatmap

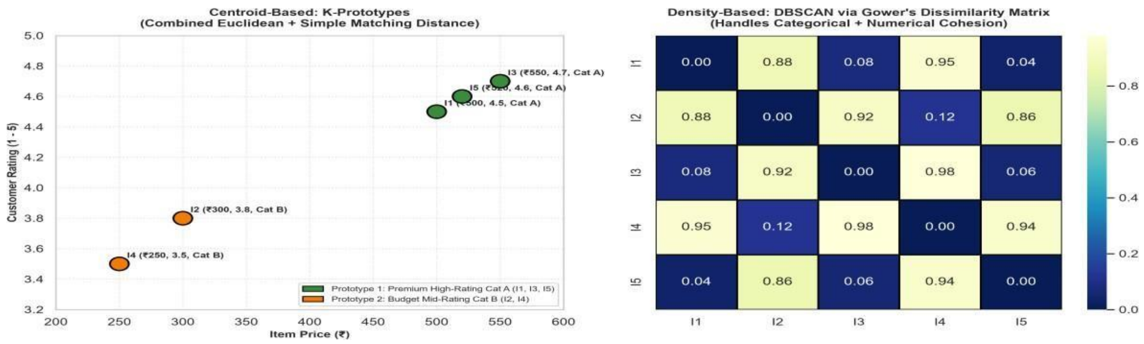


Figure 3.5: (Left) K-Prototypes centroid clustering on mixed Price and Rating; (Right) Pairwise Gower's Dissimilarity Heatmap verifying dense cluster cohesion for DBSCAN.